

PreCrash systems

PreCrash systems can harden the breaks, tighten the seatbelt, and prepare the airbags, so that the approaching crash causes minimal injury

GPS tracking

GPS tracking can be infinitely useful in plotting a route to your destination, avoiding traffic jams and even recovering your car if it gets stolen



The Toyota RAV4 offers an enviable blend of power, economy and utility thanks to its Dual VVT-i engine

technology can be seen even in some of the cheapest toy-cars.

The energy required to run a car is considerable, and unlike a toy-car, an actual car needs to run longer, faster, and needs to carry more weight. The only way to do this is to add more batteries to the car, which has the unfortunate consequence of increasing the car's weight and thus requiring even more batteries to run! With the lead-acid batteries found in cars today, making an electric car is quite inefficient.

These, if used, will also need quite some time to charge and will need to be replaced occasionally. Using NiMH or Lithium-ion cells increases the range and life of the car; at a cost – a price 10 to 20 times as much!

Electric cars can also take advantage of concepts such as regenerative braking, where energy released by braking the car is used to help recharge the batteries. Energy obtained from driving the car on a downward slope can also be used to recharge the car. These technologies greatly increase the efficiency of the car; since the energy used here is otherwise wasted.

So the only thing that stands in the way of electric cars is the battery-pack. Since improvements in

batteries affect a wide range of devices, we are in luck, as most manufacturers are already hard at work trying to improve the life and recharge characteristics of batteries, and are busy driving down their cost.

Fuel cells

Fuel cells to the rescue! These marvellous devices are capable of producing electricity directly from the fuel without involving any messy combustion. The basic concept used here is the same as an electric car;

however instead of needing to recharge the car from an external source, the fuel cell produces the required electricity. As they can be refuelled instantly, no long periods of recharging are required. Fuel cells remove most of the disadvantages of electric cars.

Additionally, since they are essentially electric cars, they too can utilise regenerative braking and energy from driving on downward slopes. They remove the long periods of recharge, increase the energy density, and don't need occasional replacements. However the cost factor remains an inhibitor.

Fuel cells have already started to become a reality for consumer electronics; however cars represent

a bigger challenge due to their much greater energy requirements. As such, fuel cell cars are currently not production ready; however, with research being done by the FreedomCAR initiative, and CaFCP (California Fuel Cell Partnership), the performance and cost issues are sure to be resolved.

For the purpose of cars, the most commonly considered option is a hydrogen fuel cell. These run on hydrogen gas and produce only water as output, making them very environmentally safe. Even if an organic compound of hydrogen such as methane or ethanol is used, there are minimal by products.

Hybrid solutions

Electric cars are certainly wonderful, but when you're unable to pick up grandma because you forgot to charge the car last night, things get embarrassing. One important consideration with any new

technology is how compatible it is with the status quo. Electric cars are suddenly alien to most of our notions of a car.

Hybrid cars are a middle ground. As their name suggests, they form an evolutionary transition between electric- and fuel-powered vehicles. They typically contain a fuel engine along with electric motors. To make the concept even more attractive, the engine in the car could be additionally made to run on alternative fuels such as CNG/LPG, decreasing the environmental impact even further. They manage to give a much better mileage, and are also capable of using regenerative braking to their advantage. Since we have essentially two engines in the same car, the gasoline engine can be made smaller than otherwise required in a normal car of similar size.

Hybrid cars can either use the fuel engine exclusively, or can run on battery power, enabling the car to be charged and run for short distance on battery power alone. Hybrid cars are becoming quite popular nowadays, as they get cheaper and are more easily available. Government initiatives also exist in some countries for people who use hybrid cars.

Alternative fuels

Most people seem to want to defer the usage of alternative fuels, until the time when there is No Other Alternative. However, gasoline's end times are approaching, and the sooner the transition to the alternatives, the better. Delhi is already a front runner in this effort, with all public transportation already running on CNG.

AIR ENGINE

Internal combustion engines rely on the expansion of the exhaust gases produced after the air-fuel mixture is combusted. Air engines instead use the energy stored in the compressed air to run the engine. Many different mechanisms are in development, from turbines to rotary motors – no avenue has been left untouched for this cleanest of energy forms. The basic principle involves using the energy from the expanding air to drive the motion of the piston, or rotor or fin. This energy is relayed to the wheels to drive the car. Since compressed air contains far lesser energy than that produced from combustion, the car needs to be optimized with lower weight and drag.